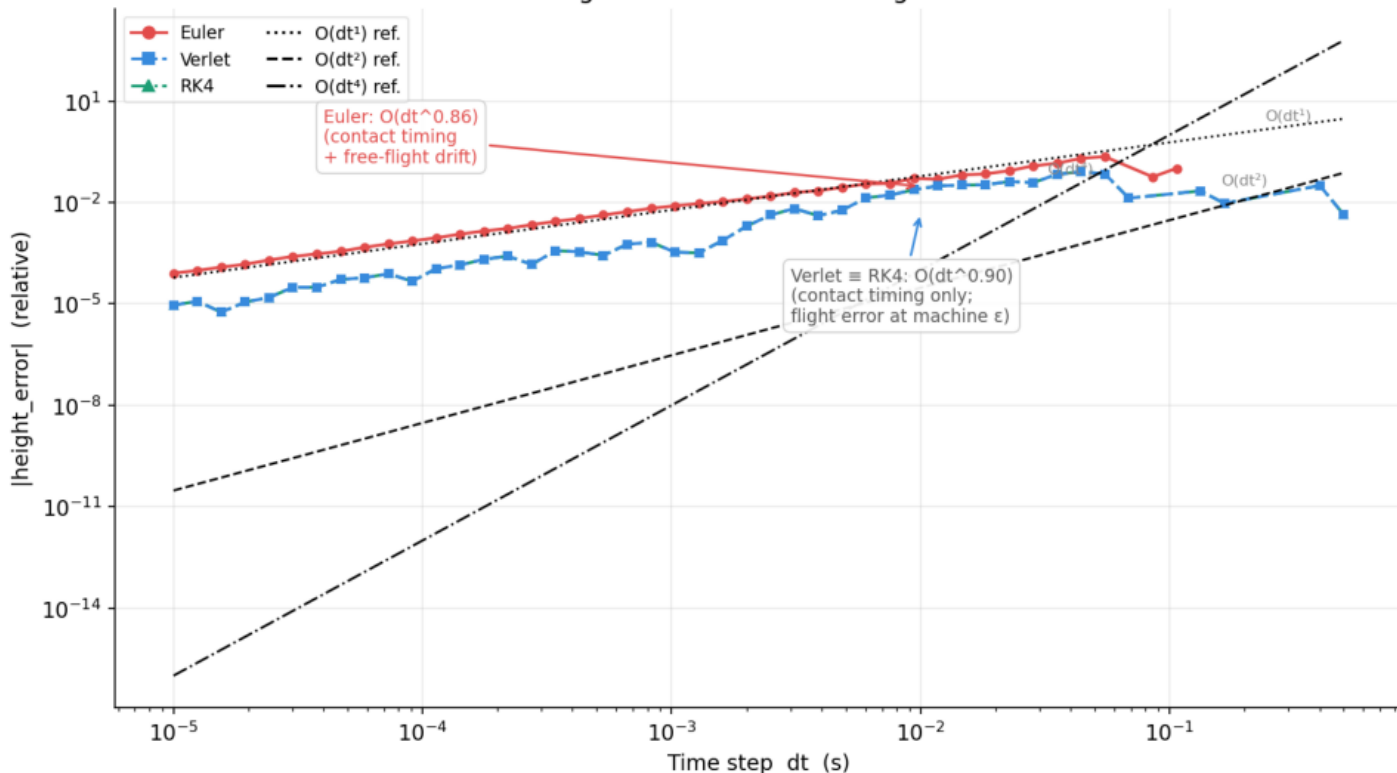


Peak height error vs dt — convergence order



CONVERGENCE ORDER

Measured slopes (log-log fit):

Euler $\rightarrow O(dt^{0.86})$ $R^2=0.978$

Verlet $\rightarrow O(dt^{0.90})$ $R^2=0.870$

RK4 $\rightarrow O(dt^{0.90})$ $R^2=0.870$

All $\sim O(dt)$ — NOT $O(dt)$, $O(dt^2)$, $O(dt^4)$ as theory predicts.

Root cause: discrete contact detection imposes an $O(dt)$ floor on all integrators.

$$v_{\text{sim}} = v^* + g \cdot (dt - \tau) = v^* + O(dt)$$

$\tau \in (0, dt)$: unknown contact frac.

Verlet $\equiv$ RK4 (identical curves):

Free flight is quadratic $\rightarrow$

both exact $\rightarrow$ same state at

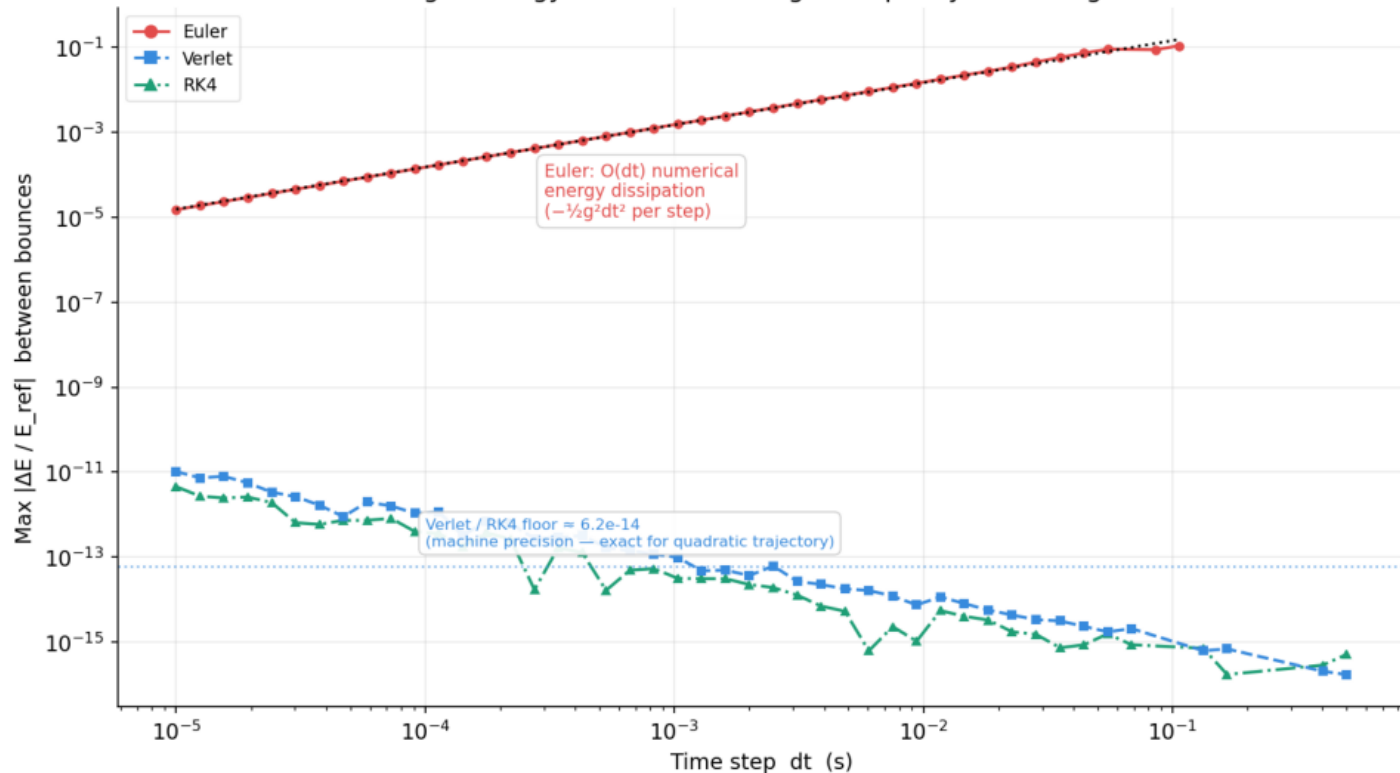
contact $\rightarrow$ identical error.

Euler 8.6 $\times$ higher than Verlet/RK4:

Extra $\sim \frac{1}{2}g^2dt^2$ dissipation/step.

$\rightarrow$ Use Figure 2 to compare integrator quality in isolation.

Flight energy drift vs dt — integrator quality in free flight



FLIGHT ENERGY DRIFT

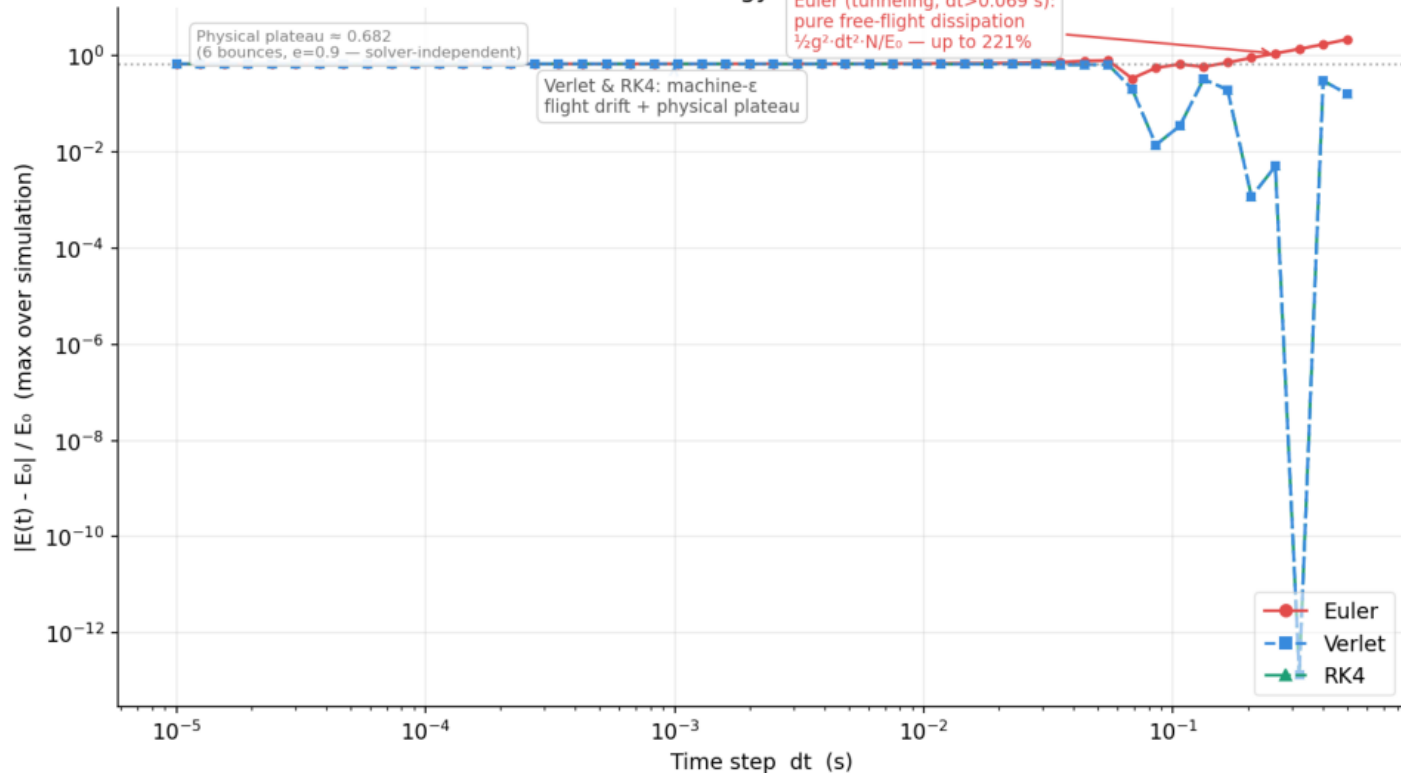
Measures integrator error only. Physical energy loss at bounces is excluded (reference reset after each bounce).

For constant-gravity free fall (quadratic trajectory):

Euler	→ $O(dt)$	drift
Verlet	→ exact	(machine ϵ)
RK4	→ exact	(machine ϵ)

Conclusion: Verlet is the best choice — same flight accuracy as RK4 at $\frac{1}{4}$ the cost.

Max energy drift vs dt



ENERGY CONSERVATION

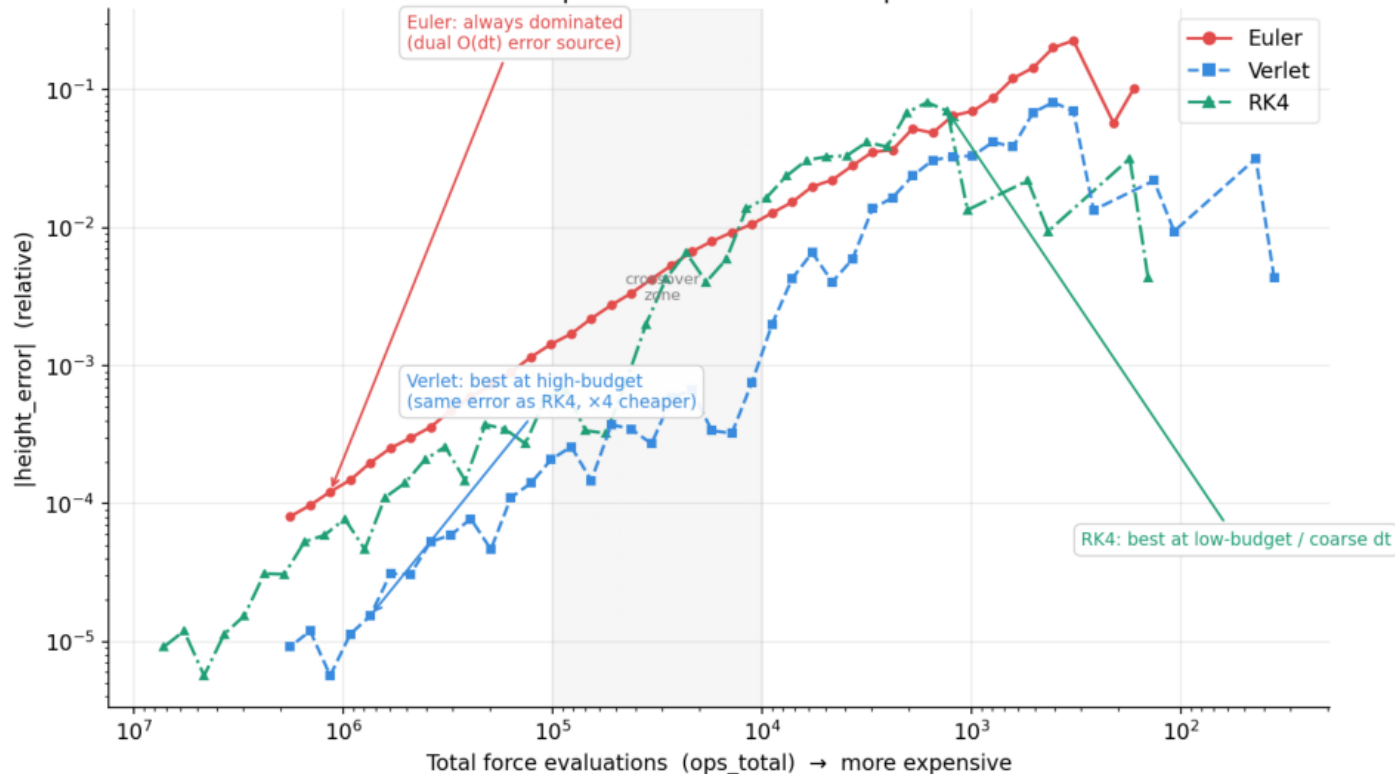
Two distinct regimes:

Large dt — Euler only ($dt > 0.069$ s):
 Ball tunnels — no bounces.
 Drift = pure Euler dissipation:
 $\frac{1}{2}g^2 \cdot dt^2 \cdot N / E_0 \rightarrow$ up to 221%
 (matches formula to 4 d.p.)
 Verlet/RK4: machine ϵ (exact free-fall, no bounce losses).

Small dt — all solvers plateau:
 max energy_drift $\approx 68.2\%$
 = physical loss from 6 bounces
 with $e=0.9$ (analytical $\approx 64.7\%$;
 gap = contact timing impulse).

All three solvers reach the
 SAME plateau $\rightarrow$ energy floor
 is set by the impulse model,
 not the integrator.

Cost vs precision — iso-cost comparison



COST vs PRECISION

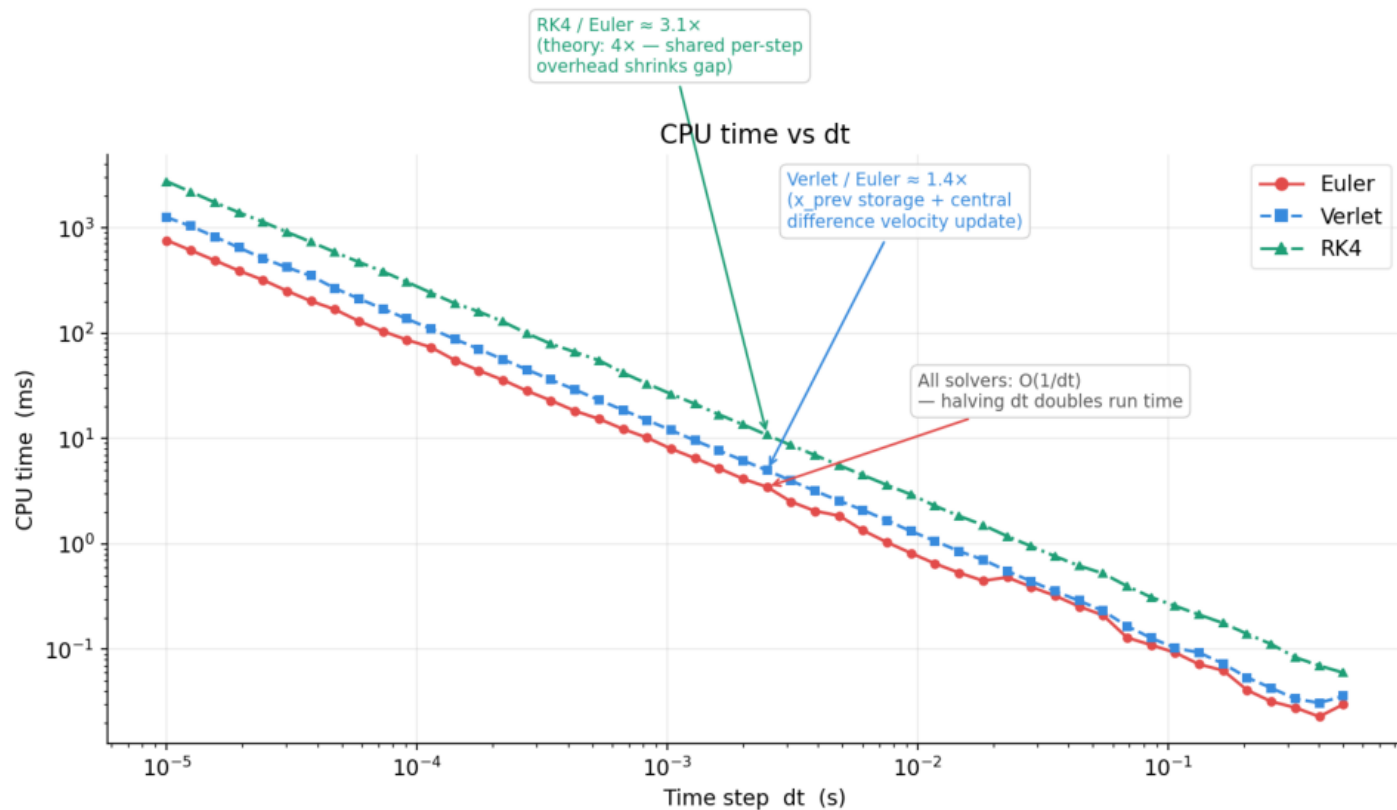
X-axis: total force evaluations
(Euler=1/step, Verlet=1/step, RK4=4/step) ← left = costlier

Euler: always dominated.
At any budget, Verlet or RK4 gives a much smaller error.

Verlet vs RK4:
Same $O(dt)$ height error (contact-timing-limited).
Verlet costs $\frac{1}{4}$ of RK4/step
→ identical precision at $4\times$ fewer ops → Verlet dominates.

At 60 Hz (120 ops/s, Verlet):
Verlet is the clear winner.

RK4 only wins at very coarse dt (low-budget, rightmost region).
Never optimal with discrete impulses – use CCD or spring forces to expose $O(dt^4)$.



CPU TIME

All solvers scale as $O(1/dt)$: parallel lines of slope -1 on this log-log plot. Halving dt doubles CPU time.

Measured cost ratios vs Euler at median dt:

Verlet : $\sim 1.44\times$

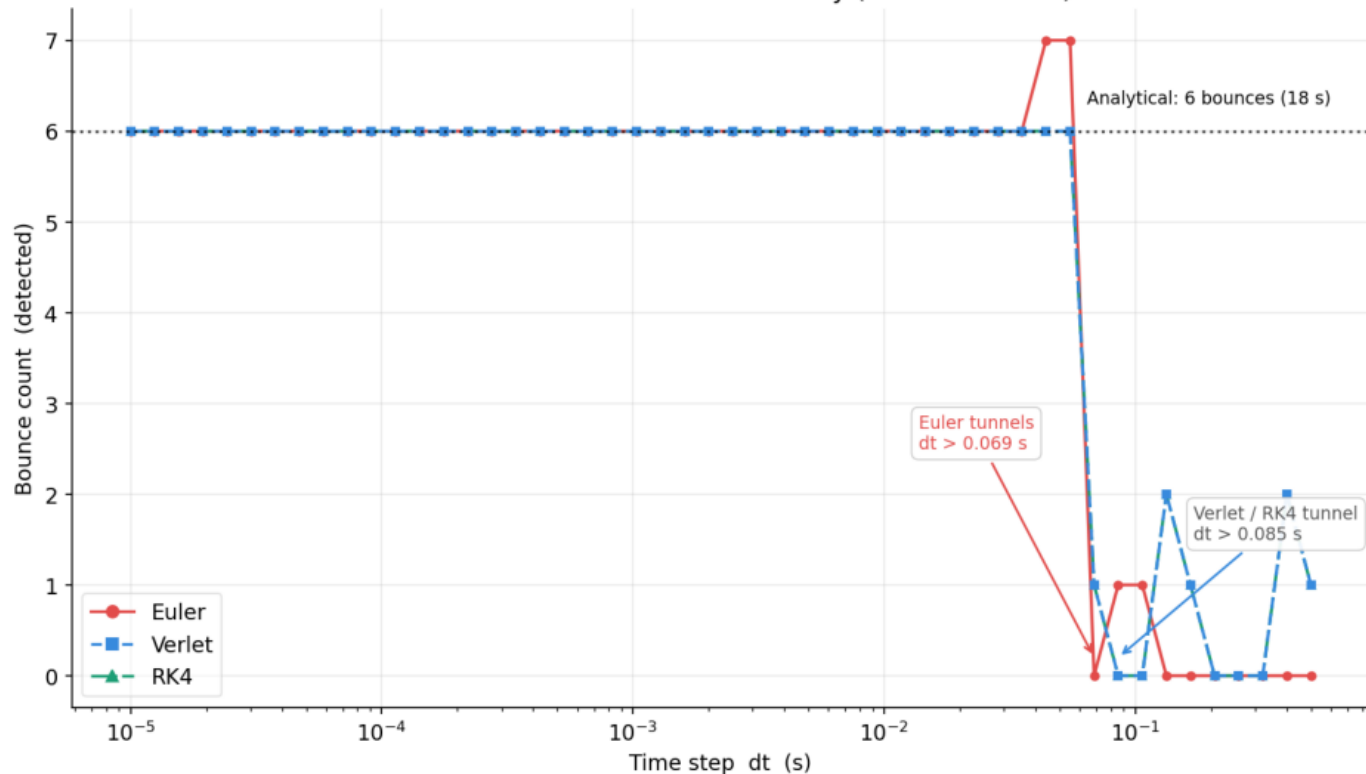
RK4 : $\sim 3.11\times$

RK4 is theoretically $4\times$ (4 force evals/step) but only measures at $\sim 3.1\times$ because per-step overhead (collision check, object lookup) is shared by all solvers.

Verlet's $\sim 1.4\times$ overhead: x_{prev} storage + central difference velocity update.

Recommendation: use Verlet. Same accuracy as RK4 at $\sim 2.2\times$ lower CPU cost.

Bounce count vs dt — solver stability (18 s simulation)



BOUNCE COUNT STABILITY

Analytical: 6 bounces in 18 s.
7th bounce at $t \approx 18.07$ s – outside the window.

Zone 1 – Tunneling (large dt):

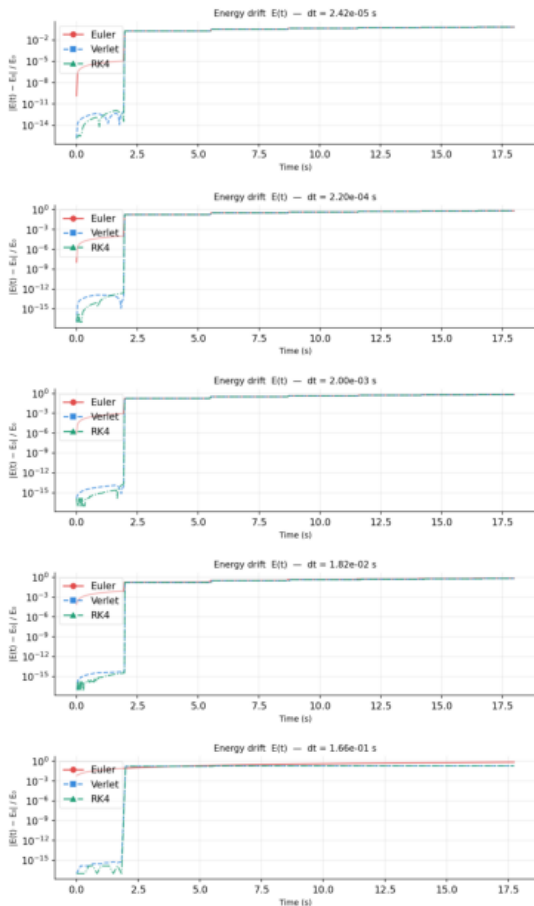
Euler : $dt > 0.069$ s $\rightarrow$ 0 bounces
Verlet/RK4: $dt > 0.085$ s
Both thresholds similar:
governed by approach velocity
at contact ($v=18.8$ m/s), not
integrator order.

Zone 2 – Stable (small dt):

All solvers: 6 bounces.
Matches analytical exactly.
Euler has 7 bounces for 2 dt
values (residual timing offset).

Zone 3 – Euler transition:

Unstable (1–5 bounces) between
tunneling and stable zones.
Verlet/RK4: sharper transition.

ENERGY DRIFT $E(t)$

Each panel shows $|E(t) - E_0|/E_0$ over the full simulation.
 E_0 = initial kinetic + potential.

Staircase jumps = physical energy loss at each bounce
 $(1 - e^2) = 19\%$ per bounce,
 $e = 0.9$). Same for all solvers.

Between bounces (free flight):
 Euler → slow upward slope
 (→ h^2/dt^2 loss per step,
 accumulates as $O(dt)$)
 Verlet = flat (exact for
 quadratic traj.) → machine ϵ
 RK4 = flat (same reason)

Coarsest dt (1.66e-01 s):
 Ball tunnels — no bounces.
 Only Euler linear drift
 is visible.

Time-domain view of the
 $O(dt)$ vs machine- ϵ gap
 quantified in Figure 2.